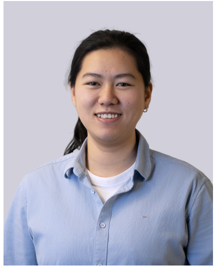
Nan Li

(+41)076-814-2673 | nan.li@irm.uzh.ch
linan1109.github.io | linan1109 | linan1109
0009-0001-7179-0746 | Nan-Li-301
Andreasstrasse 15, 8050 Zürich, Switzerland



Education

PhD in Data Science — Oct. 2024 - Current
University of Zurich (UZH) — Zurich, Switzerland

- **Research Keywords**: Traffic Medicine, Machine Learning, Human-Computer Interaction, Explainable Artificial Intelligence, Eye-tracking, EEG, Information Visualization

MSc in Computer Science (Major: Artificial Intelligence) — Sep. 2022 - Oct. 2024
University of Zurich (UZH) — Zurich, Switzerland

- **Research Keywords**: Explainable Artificial Intelligence, Human-Computer Interaction, Time-Series Classification
- **Master Thesis**: Interpretable Machine Learning Algorithm for Drunk Driving Detection
- **Master Project**: Explainability method for Recommender Systems

BEng in Software Engineering — Sep. 2016 - Jun. 2021
Beihang University (BUAA) — Beijing, China

- **GPA**: 87/100 (Top 17%)
- **Thesis**: Project Review Assistant System Based on BlockChain

Research Interests

My research focuses on leveraging machine learning algorithms to address real-world challenges. I am broadly interested in the fields of Explainable Machine Learning (**XAI**), Human-Computer Interaction (**HCI**), and **Human-Centric Software Engineering**. Currently, my work involves statistical modeling and **multi-modal time-series** analysis for **traffic safety**, with a specific emphasis on enhancing model interpretability to provide intuitive and actionable insights.

Publications & Preprints

[1] **Nan Li**, Fan Shi, Xiaoyu Zhang, and April Yi Wang. *MRISA: A Visual Analytics Approach for Comparing Locomotion Policies in Robotics Training.* In Proceedings of Graphics Interface 2025 (in press). `https://doi.org/10.1145/3769872.3769898`

[2] **Nan Li**, Veronika Gambin, Boris Quednow, Pascal Burger, Ana Buadze, Benjamin Offenhammer, Kristina Keller, and Stefan Lakämper. *Impact of drug holidays on simulated driving performance in adults with ADHD: a study protocol for a randomized crossover trial.* Under review at Trials. `http://dx.doi.org/10.21203/rs.3.rs-8611309/v1`

[3] Veronika Gambin, **Nan Li**, Esther Irene Schwarz, Kristina Keller, and Stefan Lakämper. *Validating a novel driving simulation-based MWT against the standard MWT in an OSA-cohort challenged by CPAP-withdrawal (DS-MWT2)-Protocol for a monocentric, controlled, randomized, crossover trial.* Under review at PLOS One. `http://dx.doi.org/10.64898/2026.01.18.26344362`

[4] Tzvetan Popov, **Nan Li**, Veronika Gambin, Kristina Keller, Samuel Wehrli, and Stefan Lakämper. *Predicting driver distraction using a single channel ear EEG.* Available as preprint. `https://doi.org/10.64898/2026.01.24.701469`

[5] Tzvetan Popov, **Nan Li**, Veronika Gambin, Kristina Keller, Samuel Wehrli, and Stefan Lakämper. *The neuro-ocular costs of texting during driving.* Available as preprint. `http://dx.doi.org/10.1101/2025.08.15.670515`

Awards and Honors

GRC Travel Grant from UZH	Fund	2026
Outstanding Graduate of Beihang University	Honor	Jun. 2021
Excellent Scholarship for Academic Performance (3 times)	Honor	2020, 2019, 2018
Excellent Scholarship for Social Work (2 times)	Scholarship	2019, 2018
Merit Student (2 times)	Honor	2019, 2017
Lee Kum Kee Innovation Scholarship	Scholarship	Jan. 2019

Representative Projects

Interdisciplinary Traffic Medicine and ADHD Drug Holidays
Oct. 2024 - Current

Traffic Medicine - Institute of Forensic Medicine - UZH — Zurich, Switzerland

- **Objective**: This research aimed to bridge critical gaps in the understanding of ADHD stimulant medication holidays and their effects on driving safety, enabling evidence-based, informed decision-making for patient care.
- **Contributions**:
  - Machine Learning: Utilize time-series machine learning models with driving data, eye-tracking, and EEG data to study the interaction of cognitive states and driving behavior.
  - Neuroscience: Investigate the neurocognitive effects of stimulant medication holidays on driving performance, including correlations with stimulant levels in biological samples and psychometric outcomes.
  - HCI: Design and implement a visualization platform to make machine learning decisions more intuitive and explainable, facilitating understanding for both analysis and non-specialist audiences.

Multi-Robot Interactive Simulation and Analysis Platform
Mar. 2024 - Nov.2024

Programming, Education, and Computer-Human Interaction Lab (PEACH Lab) - ETHz — Zurich, Switzerland

- **Objective**: An interactive simulation and visualization framework to facilitate quadrupedal locomotion learning.
- **Contributions**:
  - Formative Study: Conducted interviews with robotics experts to identify key challenges and strategies in locomotion policy analysis.
  - Framework Implementation: Designed and developed the framework, an interactive tool integrating simulation, trajectory, metrics visualization, and key frame capturing for locomotion policy analysis.
  - User Evaluation: Conducted a user study with robotics practitioners to assess the tool's effectiveness, demonstrating its ability to provide immediate insights and enhance exploratory analysis.

Interpretable Machine Learning Algorithm for Drunk Driving Detection
Dec. 2023 - Aug.2024

Bosch IoT Lab | ETH Zurich | University of St. Gallen — Zurich, Switzerland

- **Objective**: Develop interpretable machine learning algorithms to detect drunk driving using CAN bus data.
- **Contributions**:
  - Analyzed CAN bus and other related data from real vehicles to create accurate models.
  - Developed and trained interpretable time-series classification models including logistic regression, CNNs, RNNs, and state-of-art multivariate time-series classification models.
  - Explored techniques for comprehensive explanations of model predictions.

Explainability Method for Recommender Systems
Feb. 2023 - May. 2024

Dynamic and Distributed Information Systems Group - UZH — Zurich, Switzerland

- **Objective**: Research and implement explainable methods for recommender system.
- **Contributions**:
  - Extended the Cornac Python package with new explainable recommend models, explanation algorithms, and evaluation metrics.
  - Developed a pipeline to facilitate the workflow and documented a detailed report.
  - Specialized in matrix factorization-based models and explanation methods.

Project Review Assistant System Based on Block Chain
Nov. 2020 - Jun. 2021

School of Software - BUAA — Beijing, China

- **Objective**: Develop a project review web application utilizing blockchain technology and distributed storage.
- **Contributions**:
  - Developed the frontend using JavaScript and TypeScript with React and backend with Java and Kotlin using SpringBoot, and MongoDB.
  - Integrated blockchain technology using Hyperledger Fabric and Go-lang, with IPFS for distributed storage.
  - Enabled functionalities such as project uploads, reviews, ratings, and reviewer suggestions.

Teaching Experience

Programming in Biology	Tutor	Fall 2025
Foundations of Data Science (Graduate)	Tutor	Fall 2023
Compiler Theory (Undergraduate)	Teaching Assistant	Fall 2020
Object-Oriented Programming (Java) (Undergraduate)	Lead Teaching Assistant	Spring 2020
Algorithm Analysis and Design (Undergraduate)	Teaching Assistant	Fall 2020